

Analysis Methods for AI Evaluation

Session 6

2026-10-06

The design sets the analysis

- Random assignment makes the difference in means an unbiased estimate.
  - Pretreatment information (blocks, pairs, covariates) can make it more precise.
- **How different would it be if we ran the study again?**
  - new questions
  - new users
  - new runs on the same questions
  - a new random assignment.
- Your design fixes which of these happen.
  - Uncertainty measures follow from it.



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Your PAP must give the complete analysis recipe: point estimate, standard error, and confidence interval, each justified by the design.



Where uncertainty comes from

Every estimate comes with a margin of error

- Your study produces a number: how much changing one piece of the Bundle helps.
- Different questions, or a different coin flip for who gets which Bundle, would give a different number.
- This section asks
  - where does that variation come from?
  - how to measure it?
  - how can we conceive of and measure “correctness”?

The running example: retrieval in a course tutor

- **System:** an LLM tutor answers course questions. **Bundle B** adds retrieval over course materials; **Bundle A** does not.
- **Sample:** $N = 100$ questions drawn from the questions students could ask.
- **Assignment:** retrieval on for $m = 50$ questions at random. $Z_i = 1$ if question i gets it.
- **Outcome:** a 0–100 score for each answer. Each question is run R times; $R = 1$ until stated otherwise.

$$Y_{ir} = 50 + \tau Z_i + a_i + e_{ir}, \qquad sd(a_i) = \sigma_a = 8, \quad sd(e_{ir}) = \sigma_e = 6, \quad \tau = 3.$$

- a_i: difficulty of question i, shared by all its runs. e_{ir}: run-to-run noise.
- **Indices:** $i = 1, \dots, N$ questions; $r = 1, \dots, R$ runs; NR scored calls.

The difference in means and its design-based variance

› **Estimate:** mean score with retrieval minus mean score without.

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0, \qquad \widehat{V}(\hat{\tau}) = \frac{s_1^2}{m} + \frac{s_0^2}{N - m}.$$

```r
fit <- tidy(difference_in_means(Y ~ Z, data = study))
```

Estimate	SE	Lower 95%	Upper 95%
2.84	2.05	-1.23	6.9

Gerber & Green (2012), *Field Experiments*, eqs. 3.6–3.7.



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› The exact variance also depends on how effects vary across questions.

» That term is unobservable, so this estimate errs on the large side.

Gerber & Green (2012), *Field Experiments*, eqs. 3.6–3.7.



Two sources of uncertainty: sampling and assignment

Sampling

- Which questions enter the study.
- Repeat: draw new questions.
- Estimand: average effect over the **population** of questions.

Assignment

- Which questions get retrieval.
- Repeat: re-randomize the same questions.
- Estimand: average effect on **these** questions.

Abadie, Athey, Imbens & Wooldridge (2023), *When Should You Adjust Standard Errors for Clustering?*, QJE 138(1).



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- The variance formula above covers both when questions are a random sample.
- A claim about new questions needs the sampling term.
  - The question set has to be a credible sample of that population.

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- The variance formula above covers both when questions are a random sample.
- A claim about new questions needs the sampling term.
  - The question set has to be a credible sample of that population.
- **Abadie et al.:** clustering is a statement about how sampling or assignment worked.

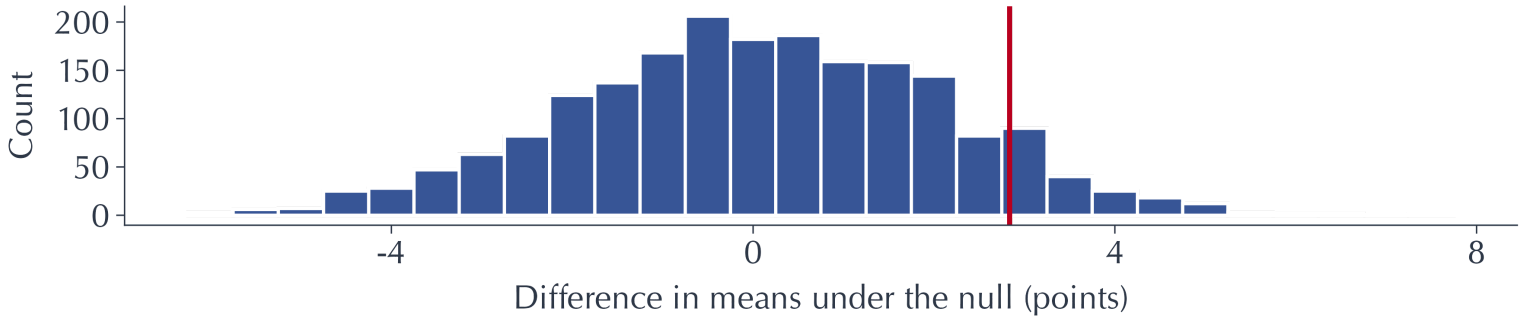
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Randomization inference

- **Sharp null:** each question scores the same with or without retrieval.
- Re-randomize the 100 questions 2,000 times; recompute the difference each time.

```r
null_diffs <- permutation_draws(study, draws = 2000)
observed <- with(study, mean(Y[Z == 1]) - mean(Y[Z == 0]))
p_value <- mean(abs(null_diffs) >= abs(observed))
```

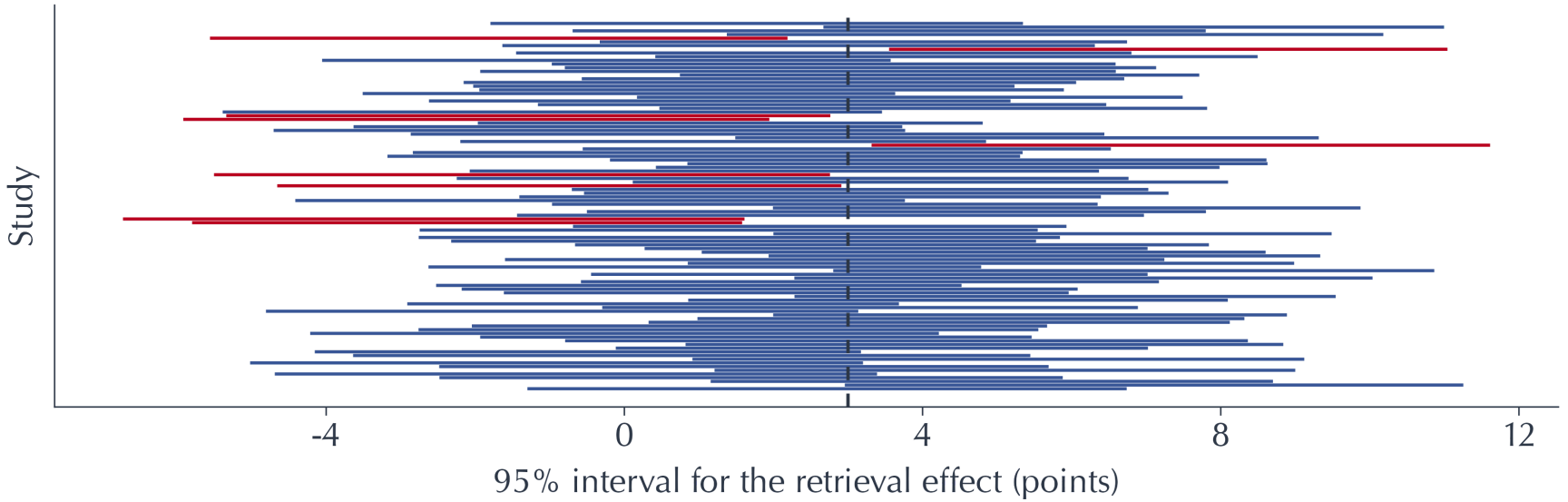


The red line is the observed difference. Two-sided $p = 0.170$.

Gerber & Green (2012), *Field Experiments*, §3.4.

Coverage is the test of a standard error

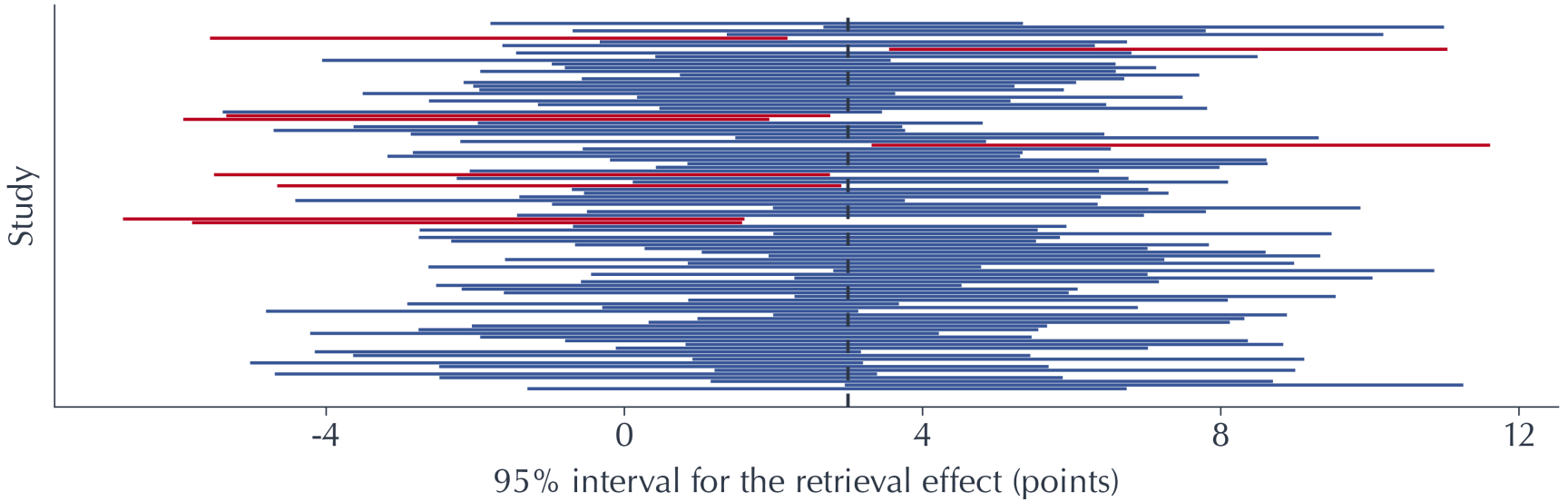
- **Coverage:** the fraction of 95% intervals containing the true effect.
- **Template:** repeat the whole design, compute each interval, count the hits.



- First 100 of 250 studies. Red intervals miss the true 3-point effect.
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- Every simulation today reports coverage for a design and a standard error.

Estimate, variance, and coverage

- Random assignment makes the **difference in means** unbiased. The **Neyman variance** gives its standard error from the design alone.
- Uncertainty has two sources: **sampling** the questions and **assigning** treatment. Which ones you count decides whether you learn about the population or about these questions.
- **Randomization inference** tests the sharp null using the assignment step alone.
- **Coverage** checks any standard error: simulate the design and see whether 95% intervals contain the truth 95% of the time.



Matching standard errors to the design

How to deal with repeated measures

- Eval data often reuses things:
  - the same question run many times
  - the same graders scoring every answer
- Measurements that share a question or a grader overlap in what they tell you.
- Counting each one as fresh evidence makes results look more certain than they are.
- This section provides tools to accurately reflect uncertainty

A clustered SE sums residuals within each question

- **Mean score** over all NR calls; residual $\hat{e}_{ir} = Y_{ir} - \bar{Y}$.

$$\widehat{V}_{\text{iid}}(\bar{Y}) = \frac{1}{(NR)^2} \sum_{i=1}^{N} \sum_{r=1}^{R} \hat{e}_{ir}^2, \qquad \widehat{V}_{\text{cluster}}(\bar{Y}) = \frac{1}{(NR)^2} \sum_{i=1}^{N} \Big(\sum_{r=1}^{R} \hat{e}_{ir} \Big)^2.$$

- Expanding the square adds $\hat{e}_{ir}\hat{e}_{is}$ for every pair of runs $r \neq s$ on the same question.
  - » In the model both residuals contain a_i, so these products average about $\sigma_a^2 > 0$.
  - » Products across questions are set to zero: questions are the independent units.
- The approximation needs many clusters: N questions, not NR calls, is the sample size.

Miller (2024), *Adding Error Bars to Evals*, eq. 4.

In regression, clustering changes the sandwich

- **Regression:** $\mathbf{Y} = \mathbf{X}\beta + \boldsymbol{e}$ by OLS. Clusters $g = 1, \ldots, G$ (here, questions); $\mathbf{X}_g$, $\hat{\boldsymbol{e}}_g$ are cluster g's rows and residuals.

$$\widehat{V}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\Big(\sum_{g=1}^{G} \mathbf{X}_g'\hat{\boldsymbol{e}}_g\hat{\boldsymbol{e}}_g'\mathbf{X}_g\Big)(\mathbf{X}'\mathbf{X})^{-1}.$$

- The outer terms are the OLS "bread". The middle "meat" treats each cluster's score $\mathbf{X}_g'\hat{\boldsymbol{e}}_g$ as one independent draw.
- One call per cluster: the meat is $\sum_i \mathbf{x}_i\mathbf{x}_i'\hat{e}_i^2$, the robust (HC0) variance.
- With $\mathbf{x}_i = (1, Z_i)$, the entry for $\hat{\tau}$ is the clustered variance of the difference in means.
- **CR2** (the `lm_robust` default): replace $\hat{\boldsymbol{e}}_g$ with $(\mathbf{I}_g - \mathbf{H}_{gg})^{-1/2}\hat{\boldsymbol{e}}_g$. $\mathbf{H}_{gg}$: cluster g's block of $\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$; $\mathbf{I}_g$: identity matrix.

Bell & McCaffrey (2002), *Bias Reduction in Standard Errors for Linear Regression with Multi-Stage Samples*, Survey Methodology 28(2).

Repeated calls are clustered within questions

- **Running example with $R = 5$:** 500 scored calls on 100 questions.
- Every run of question i shares a_i. Average the runs of each question:

$$\bar{Y}_i = 50 + \tau Z_i + a_i + \bar{e}_i, \qquad V(\hat{\tau}) = \left(\sigma_a^2 + \frac{\sigma_e^2}{R}\right)\left(\frac{1}{m} + \frac{1}{N - m}\right).$$

- More runs shrink only the run-noise term.

Kish (1965), *Survey Sampling*, Wiley.

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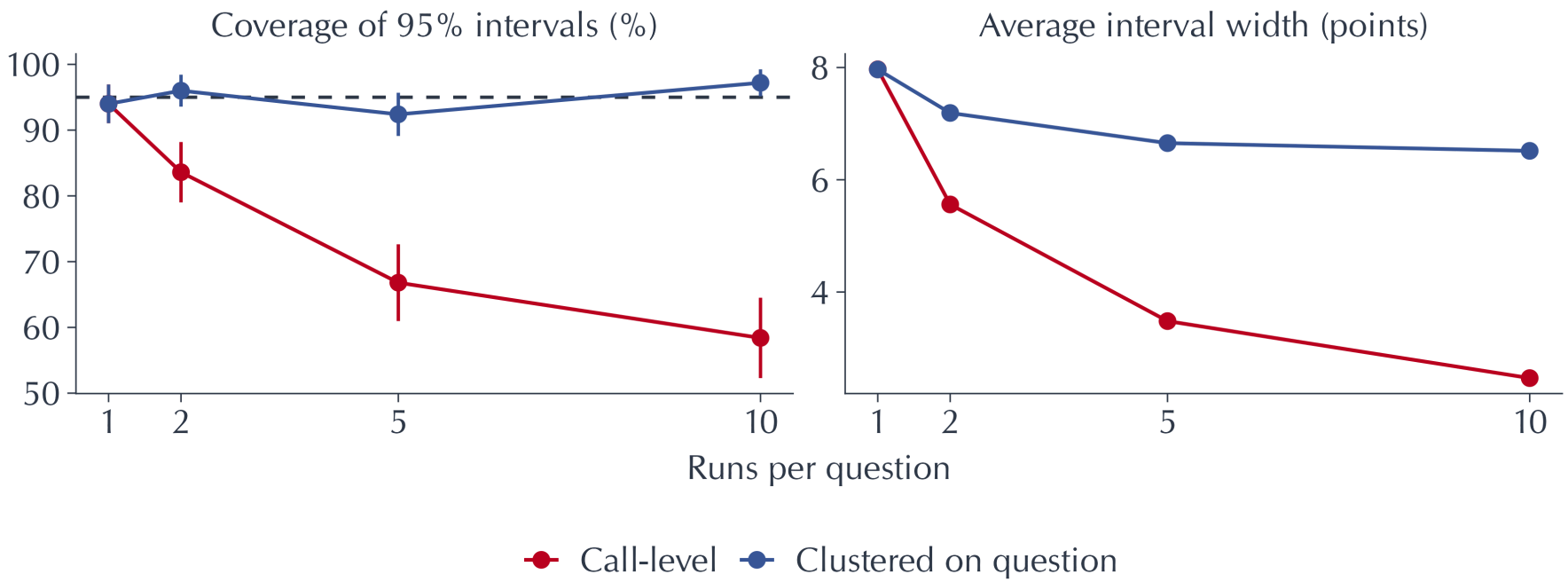
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- More runs shrink only the run-noise term.
- A call-level SE treats the NR calls as independent.
  - It uses $(\sigma_a^2 + \sigma_e^2)/R$ in place of $\sigma_a^2 + \sigma_e^2/R$.
- The ratio is the **design effect**: $1 + (R-1)\rho$, with $\rho = \sigma_a^2/(\sigma_a^2 + \sigma_e^2)$.
  - Here $\rho = 0.64$, $R = 5$: variance ratio 3.6, SE ratio **1.9**.

Kish (1965), *Survey Sampling*, Wiley.

Simulation: call-level standard errors undercover

- 100 questions; $R = 1, 2, 5, 10$ runs each; 250 studies per point.



- One run per question: the two SEs coincide.
- More runs narrow the call-level interval. The clustered interval barely moves, because question difficulty dominates.

Cluster where the design groups units

- **Abadie et al.:** the sampling process and the assignment mechanism decide whether and where to cluster.
- **Clustered assignment** (all runs of a question, a user, a classroom): cluster at the assignment level.
- **Clustered sampling** (passages, questions with repeated runs): cluster at the sampling level.

Abadie, Athey, Imbens & Wooldridge (2023), *When Should You Adjust Standard Errors for Clustering?*, QJE 138(1). Miller (2024), *Adding Error Bars to Evals*, Table 4.



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- **Neither:** questions sampled and assigned one at a time. Use robust SEs.
  - Clustering on topic widens intervals for no reason, even if scores within a topic correlate.
- **Treatment varies within sampled clusters**, as in pairs: cluster SEs are conservative.
  - The paired SE, or Abadie et al.'s CCV estimator, is tighter.

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- **Treatment varies within sampled clusters**, as in pairs: cluster SEs are conservative.
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- **Miller:** clustering reading comprehension tasks by passage makes the SE **3×** larger.

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Pairs and blocks: compare within, then average

- **Blocks** $j = 1, \ldots, J$, formed before assignment: $Y_i = 50 + \tau Z_i + \gamma_{j(i)} + \epsilon_i$.
  - Randomize within each block; block j has N_j questions, m_j treated.

$$\hat{\tau} = \sum_{j=1}^{J} \frac{N_j}{N} \hat{\tau}_j, \qquad \widehat{V}(\hat{\tau}) = \sum_{j=1}^{J} \left(\frac{N_j}{N}\right)^2 \left(\frac{s_{1j}^2}{m_j} + \frac{s_{0j}^2}{N_j - m_j}\right).$$

- $\hat{\tau}_j$, the difference in means within block j, cancels the shared γ_j.

Gerber & Green (2012), *Field Experiments*, §§3.6.1, 4.4. Miller (2024), *Adding Error Bars to Evals*, §4.2.



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- **Pairs** are blocks of two: question j runs once per Bundle, so $\hat{\tau}_j = D_j = Y_{jB} - Y_{jA}$.
  - One unit per arm leaves s_{1j}^2 undefined; use the spread of the D_j:

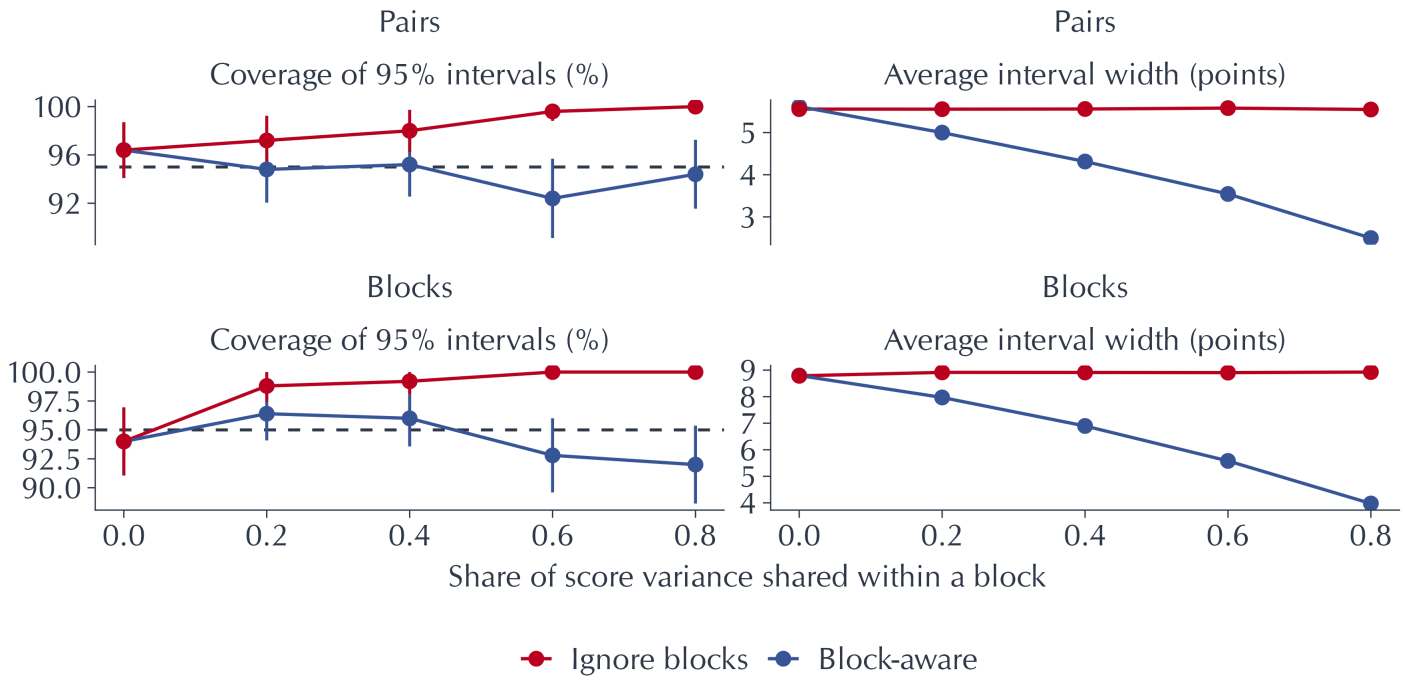
$$\widehat{V}(\hat{\tau}) = \frac{s_D^2}{J}, \qquad s_D^2 = s_A^2 + s_B^2 - 2s_{AB}.$$

- Ignoring the pairs drops s_{AB}.

Gerber & Green (2012), *Field Experiments*, §§3.6.1, 4.4. Miller (2024), *Adding Error Bars to Evals*, §4.2.

Simulation: ignoring pairs or blocks

› **Pairs:** both Bundles once on 100 questions. **Blocks:** lecture 05 design, 80 questions in 4 difficulty blocks.



› Same $\hat{\tau}$ either way here. Ignoring the blocks keeps their variation in the SE: intervals stay wide.

Questions and annotators are both sampled

› **Human-scored running example:** the same $K = 20$ annotators, drawn from a pool of graders, score all N answers.

$$Y_{ik} = 50 + \tau Z_i + a_i + b_k + \gamma_k Z_i + \varepsilon_{ik}.$$

› a_i: difficulty. b_k: leniency. γ_k: how much annotator k rewards retrieval answers.

Westfall, Judd & Kenny (2015), *Replicating Studies in Which Samples of Participants Respond to Samples of Stimuli*, Perspectives on Psychological Science 10(3).



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- **Target:** the effect as judged by new annotators on new questions.

$$V(\widehat{\tau}) \approx \frac{4(\sigma_a^2 + \sigma_\varepsilon^2/K)}{N} + \frac{\sigma_\gamma^2}{K}.$$

- b_k cancels: every annotator scores both arms. The γ_k term depends only on the 20 annotators.

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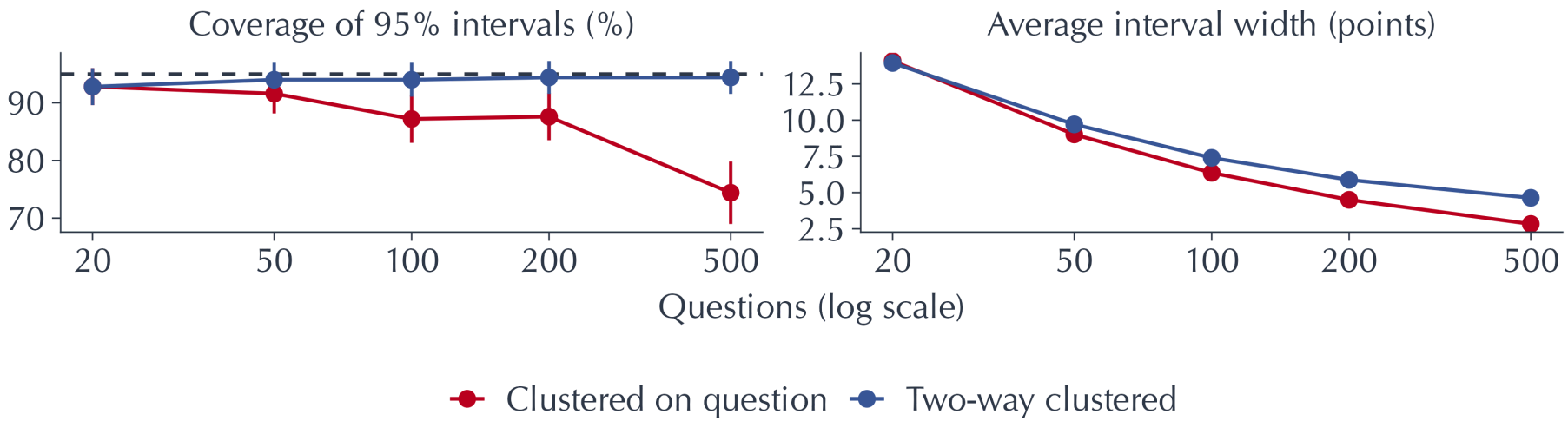
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- **Westfall et al.:** with two crossed samples, power hits a ceiling as one grows. The other's size sets it.

Westfall, Judd & Kenny (2015), *Replicating Studies in Which Samples of Participants Respond to Samples of Stimuli*, Perspectives on Psychological Science 10(3).

Simulation: one-way versus two-way clustering

- 20 annotators; 20 to 500 questions. $\sigma_a = 8$, $\sigma_b = 6$, $\sigma_\gamma = 4$, $\sigma_\varepsilon = 6$.
- **Two-way:** $\widehat{V} = \widehat{V}_{question} + \widehat{V}_{annotator} - \widehat{V}_{score}$ (sandwiches); t_{19}.

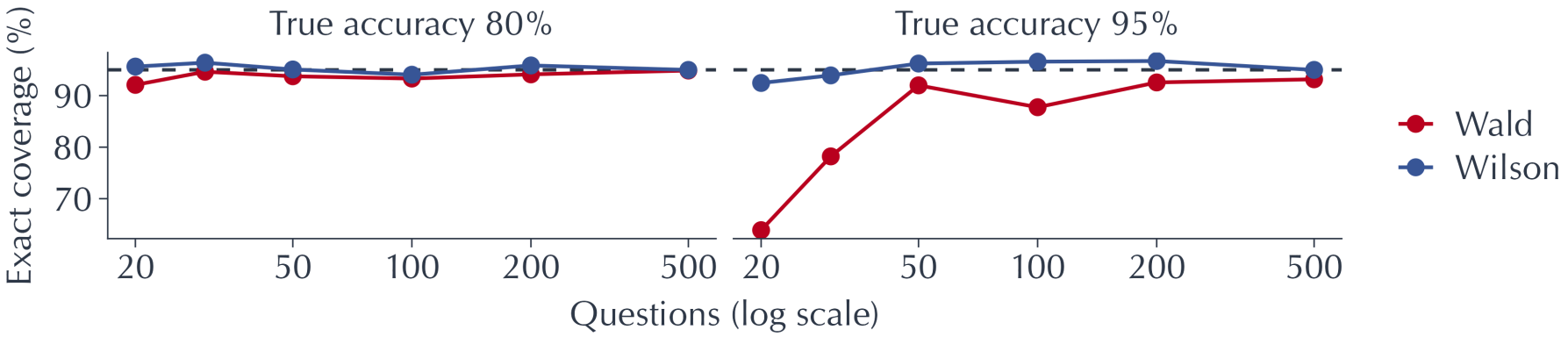


- Question-only clustering treats the 20 annotators as fixed. Coverage falls as questions are added.

Cameron, Gelbach & Miller (2011), *Robust Inference With Multiway Clustering*, JBES 29(2).

Small samples break the normal approximation

$$\text{Wald: } \hat{p} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{N}}, \qquad \text{Wilson: } \left\{p : |\hat{p} - p| \le z\sqrt{\frac{p(1-p)}{N}}\right\}.$$



- $\hat{p} = x/N$, $z = 1.96$. Wald plugs in $\hat{p}$, so 20 of 20 gives $[1, 1]$. Wilson uses each candidate p.
- 95% accuracy, 20 questions: Wald covers 64%, Wilson 92%.

Bowyer, Aitchison & Ivanova (2025), ICML. Wilson (1927), JASA 22(158).

Benchmark scores depend on the questions

- A benchmark score averages over the questions its authors chose.
  - Its SE assumes they are a random sample from the population you care about.

Ailem, Marazopoulou, Siska & Bono (2024), *Examining the Robustness of LLM Evaluation to the Distributional Assumptions of Benchmarks*, ACL. Perlitz et al. (2024), *Efficient Benchmarking (of Language Models)*, NAACL.



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- **Ailem et al.:** model performance is correlated across prompts.
  - Correlations track semantic similarity and shared failure points.
  - Accounting for them changes model rankings on major benchmarks.
- **Perlitz et al.:** HELM rankings shift with benchmark design choices.
  - Removing one low-ranked model can change the leader.

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  - Removing one low-ranked model can change the leader.
- **In your PAP:**
  - define the question population and how questions are drawn from it
  - cluster where questions arrive in groups
  - report the effect by topic as a secondary analysis.

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Or is a benchmark just a fixed exam?

- An error bar asks how the score would move with a fresh draw of questions.
- A benchmark's questions were chosen, curated, and frozen.
  - There is no population they were drawn from, so the error bar describes an imagined resampling.
- **Recht:** benchmarks work as a *shared competition*. Everyone takes the same exam, and results are reproducible.

Recht (2025), *Standard error of what now?*, arg min. Recht (2025), *Benchmarking our benchmarks*, arg min. Recht, Roelofs, Schmidt & Shankar (2019), *Do ImageNet Classifiers Generalize to ImageNet?*, ICML.



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  - That drop is far outside any error bar on the original test set.
  - The *ranking of models* was almost exactly preserved.

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  - The *ranking of models* was almost exactly preserved.
- Uncertainty still matters because deployment is a kind of replication.

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Standard errors that match the design

- The standard error must treat as independent exactly the units the design sampled or assigned independently.
- Repeated runs share a question, so **cluster** on the question. The **design effect** says how much call-level SEs understate, and it grows with more runs.
- **Pairs and blocks** are compared within, so their SEs drop the shared variation.
- Sampled questions and sampled annotators are **crossed**. **Two-way clustering** counts both, and the smaller sample caps precision.
- With few units the normal approximation fails. Use **small-sample intervals**.



Covariates buy precision

Use what you know before the experiment

- Before the study runs, you often know something about each question: its topic, how hard it was in a pilot.
- Retrieval cannot change that information, because it was recorded first.
- Using it in the analysis sharpens the comparison and leaves the target the same.
- This section shows how to use it.

The Lin estimator: adjust with interactions

- **Freedman (2008):** adding covariates to the regression can reduce precision, and its usual SEs can be wrong.
  - The problem arises when treatment changes how covariates relate to the outcome.

Freedman (2008), *On Regression Adjustments to Experimental Data*, Advances in Applied Mathematics 40(2). Lin (2013), *Agnostic Notes on Regression Adjustments to Experimental Data*, Annals of Applied Statistics 7(1). MacKinnon & White (1985), *Some Heteroskedasticity-Consistent Covariance Matrix Estimators with Improved Finite Sample Properties*, J. Econometrics 29(3).



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  - The problem arises when treatment changes how covariates relate to the outcome.
- **Lin (2013):** center the covariate and interact it with treatment.

$$Y_i = \alpha + \tau Z_i + \beta(X_i - \bar{X}) + \gamma Z_i(X_i - \bar{X}) + \varepsilon_i.$$

- $\hat{\tau}$ estimates the average effect.
- **HC2 SE:** the sandwich with meat $\sum_i \mathbf{x}_i \mathbf{x}_i' \hat{\varepsilon}_i^2/(1 - h_{ii})$; $\mathbf{x}_i$ is row i of the regression, h_{ii} its leverage.
- In large samples it is at least as precise as the difference in means.

Freedman (2008), *On Regression Adjustments to Experimental Data*, Advances in Applied Mathematics 40(2). Lin (2013), *Agnostic Notes on Regression Adjustments to Experimental Data*, Annals of Applied Statistics 7(1). MacKinnon & White (1985), *Some Heteroskedasticity-Consistent Covariance Matrix Estimators with Improved Finite Sample Properties*, J. Econometrics 29(3).



The Lin estimator: adjust with interactions

- **Freedman (2008):** adding covariates to the regression can reduce precision, and its usual SEs can be wrong.
  - The problem arises when treatment changes how covariates relate to the outcome.
- **Lin (2013):** center the covariate and interact it with treatment.

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```
lm_lin(Y ~ Z, covariates = ~ pretest, data = study)
```

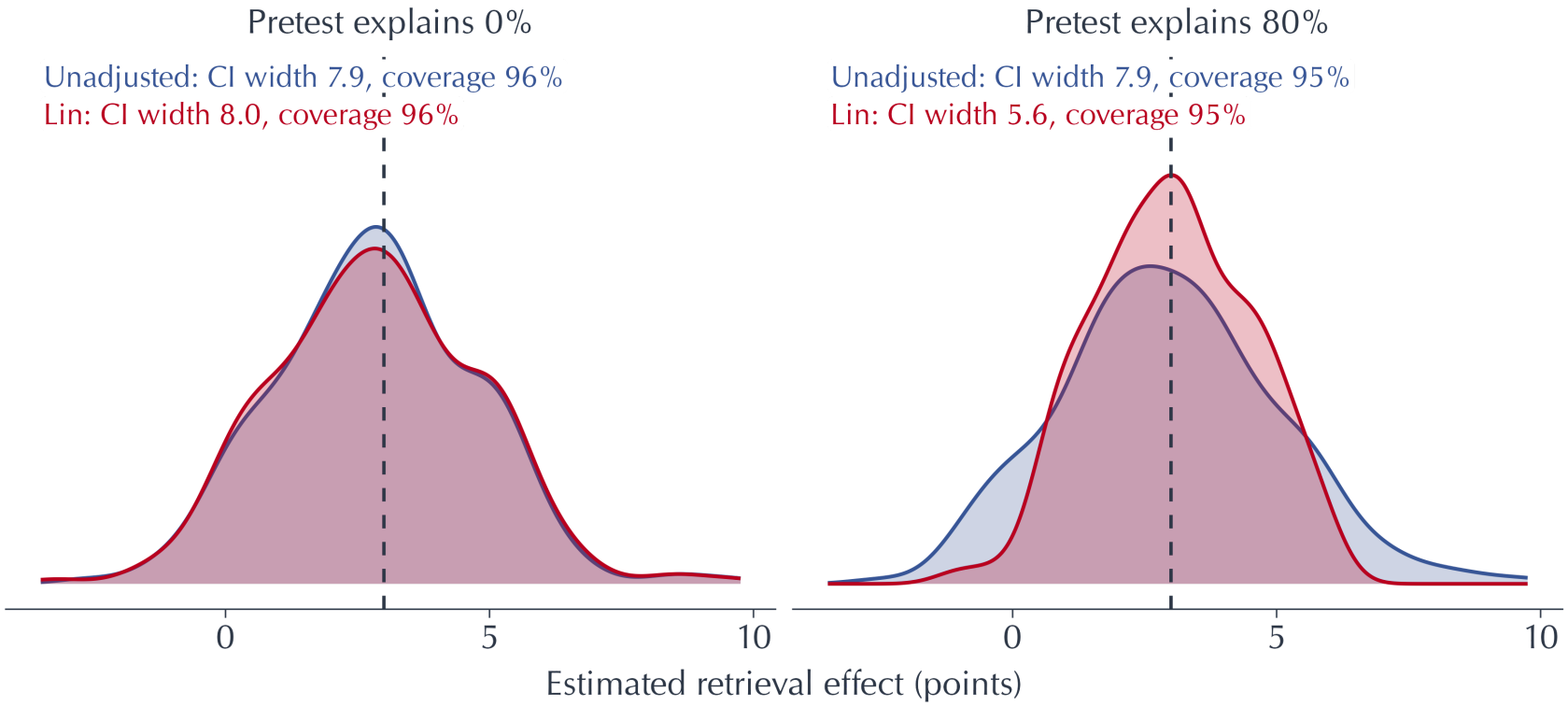
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Simulation: Lin adjustment narrows intervals

› **Covariate:** a pretreatment difficulty score for each of the 100 questions.

» It explains 0% or 80% of the variation across questions.



› Coverage stays near 95%. Width shrinks when the covariate predicts.

Covariates are fixed before outcomes are seen

- **Pretreatment:** measured before assignment, so treatment cannot change it.
  - A pilot difficulty score, question topic, a student's baseline quiz score.
- **Post-treatment:** answer length, number of retrieval calls, time spent with the tutor.
  - Adjusting for these biases the effect estimate.

Gerber & Green (2012), *Field Experiments*, ch. 4. Lin (2013), *Agnostic Notes on Regression Adjustments to Experimental Data*, Annals of Applied Statistics 7(1).



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- Choose covariates the way you chose blocks: by what predicts the outcome.
- Name them in the PAP, with the estimator.
  - Trying several sets and reporting the best gives intervals that are too narrow.

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- Blocked designs can adjust too: block indicators as covariates.

Gerber & Green (2012), *Field Experiments*, ch. 4. Lin (2013), *Agnostic Notes on Regression Adjustments to Experimental Data*, Annals of Applied Statistics 7(1).



AI predictions as covariates

- Before the experiment, a model can predict each question's score.
  - Ask an LLM how hard the question is, or score a cheap model's answer.
- **Arbour et al.:** add the prediction as a covariate in the Lin regression.

```
lm_lin(Y ~ Z, covariates = ~ pretest + ai_prediction, data = study)
```

Arbour, Ben-Michael, Feller, Lal & Yuan (2026), *AI-Assisted Variance Reduction in Randomized Experiments*.



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```

- Randomization keeps the estimate unbiased however poor the prediction is.
  - A useless prediction gets a coefficient near zero: the difference in means returns.
  - A good prediction narrows the interval.
- The prediction must use only pretreatment inputs, such as the question text.

Arbour, Ben-Michael, Feller, Lal & Yuan (2026), *AI-Assisted Variance Reduction in Randomized Experiments.*



Covariate adjustment summary

- **Covariate adjustment** buys precision. Randomization already provides unbiasedness.
- The **Lin estimator**, with treatment interactions and robust SEs, is at least as precise as the difference in means in large samples.
- The gain depends on how well the covariate predicts the outcome. A useless covariate costs almost nothing.
- Only **pretreatment** covariates qualify, named in the PAP before outcomes are seen. **AI predictions** built from pretreatment inputs count.

Measured outcomes add uncertainty

What if the scores themselves are off?

- So far we have trusted the score each answer received.
- AI judges often do some of the scoring; they make mistakes.
- Some mistakes only add noise. Others push the result in one direction.
- This section shows how a small set of human ratings lets you check the judge and correct for it.

LLM-judged outcomes are measured with error

- **Human score** H_i: the PAP's outcome. **Judge score** J_i: the LLM judge's score.

$$J_i = H_i + \delta Z_i + u_i.$$

- u_i: noise independent of treatment. It widens the interval.
- δ: error that depends on treatment. Retrieval answers cite sources; the judge rewards citations.

Egami, Hinck, Stewart & Wei (2023), *Using Imperfect Surrogates for Downstream Inference*, NeurIPS. Knox, Lucas & Cho (2022), *Testing Causal Theories with Learned Proxies*, Annual Review of Political Science.

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- The difference in means on J estimates $\tau + \delta$.
  - Its SE is correct for the effect on the judge's score.
  - More questions shrink the interval around $\tau + \delta$, so coverage of τ falls as N grows.

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- **Knox et al.:** only a held-out sample of true labels identifies the proxy's error.
- **Egami et al.:** use those true labels to calibrate how the judge's labels enter the estimate.

Egami, Hinck, Stewart & Wei (2023), *Using Imperfect Surrogates for Downstream Inference*, NeurIPS. Knox, Lucas & Cho (2022), *Testing Causal Theories with Learned Proxies*, Annual Review of Political Science.

Correct the judge with some human labels

- The judge scores all N answers. Humans score a random sample: $R_i = 1$ with known probability π, set independently of Z_i.
- **Design-based supervised learning (DSL)** pseudo-outcome (Egami et al.):

$$\tilde{Y}_i = J_i + \frac{R_i}{\pi}\big(H_i - J_i\big), \qquad \hat{\tau} = \bar{\tilde{Y}}_1 - \bar{\tilde{Y}}_0.$$

- $E[\tilde{Y}_i \mid H_i, J_i] = H_i$: the correction removes δ and all other judge error.
- Neyman SE on $\tilde{Y}$. In one arm with N answers, $n = \pi N$ of them human-labeled:

$$V\big(\bar{\tilde{Y}}\big) = \frac{V(H_i)}{N} + \Big(\frac{1}{n} - \frac{1}{N}\Big)E\big[(H_i - J_i)^2\big].$$

- Human labels alone: $V(H_i)$ replaces $E[(H_i - J_i)^2]$.
  - The judge helps when its mean squared error is below $V(H_i)$.

Egami, Hinck, Stewart & Wei (2023), *Using Imperfect Surrogates for Downstream Inference*, NeurIPS, eq. 4.

Prediction-powered inference tunes a weight

- **PPI (Boyeau et al.):** N judge scores, n human labels, and a weight λ on the judge:

$$\hat{\mu} = \frac{\lambda}{N} \sum_{i=1}^{N} J_i + \frac{1}{n} \sum_{i:R_i=1} \left(H_i - \lambda J_i \right).$$

- $\lambda = 1$ is the DSL estimate with $\pi = n/N$.
- $\lambda = 0$ uses the human labels alone.
- **PPI++** picks λ to minimize the variance, so the estimate is at least as precise as human labels alone.

Boyeau, Angelopoulos, Yosef, Malik & Jordan (2024), *AutoEval Done Right*, eq. 4.

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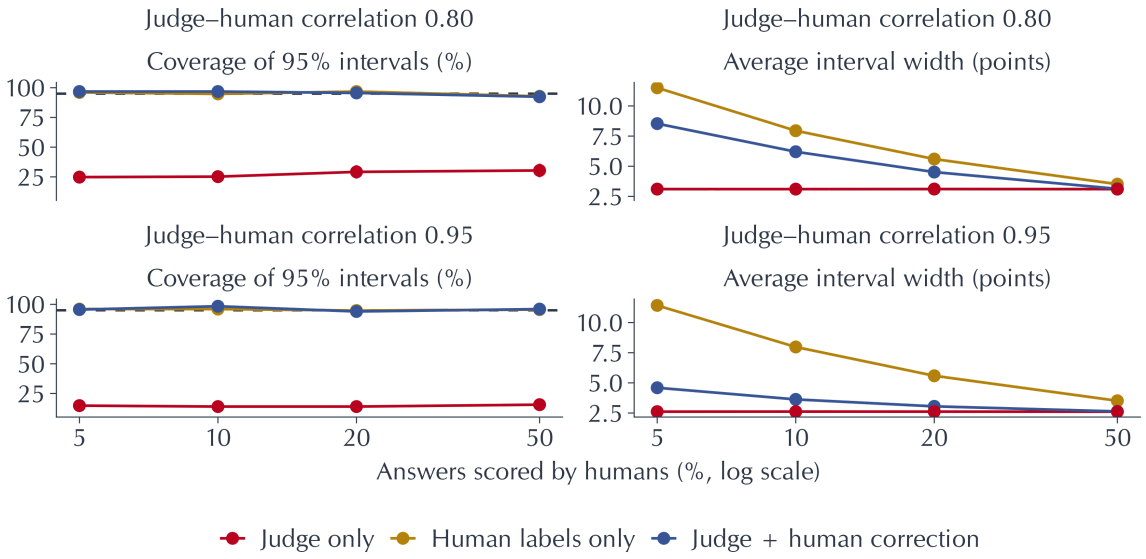
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- A poor judge makes $\mathbb{E}[(H_i - J_i)^2]$ larger than $\mathbb{V}(H_i)$. Then $\lambda = 1$ is less precise than ignoring the judge.
- For a treatment effect, apply the estimate in each arm and take the difference.

Boyeau, Angelopoulos, Yosef, Malik & Jordan (2024), *AutoEval Done Right*, eq. 4.

Simulation: judge-only versus corrected intervals

› 1,000 questions; the judge scores all and adds $\delta = 2$ to retrieval answers. Humans score a random 5–50%.



› Judge-only intervals mostly miss. The more accurate judge covers less: a narrower interval around $\tau + \delta$.

› Corrected intervals cover. With the 0.95 judge, 10% human labels match the width of 50% labeled by hand.

Correcting judge-scored outcomes

- An LLM judge is a **surrogate** for the human score. Judge-only intervals are valid for the effect on the judge's score.
- **Differential error**, where the judge favors one arm, biases the estimate. More questions make the interval more confidently wrong.
- A **random sample of human labels** with known probability identifies the judge's error. **DSL** and **PPI** use it to correct the estimate.
- The corrected interval includes the labeling uncertainty. A more accurate judge needs fewer human labels for the same precision.

Workshop preparation

Plan for missing outcomes and many tests

Missing outcomes

- Refusals, API errors, timeouts, users who stop.
- Give every assigned unit an outcome: a refusal or failed call scores as a failure.
  - Then analyze everyone as assigned.
- Dropping them compares different subsets if treatment changes who goes missing.
- Fix the retry rule in advance. Report missingness by arm.

Many tests

- k independent tests at 5%: the chance of at least one false positive is $1 - 0.95^k$, or **40%** for $k = 10$.
- Name one primary outcome. Label the rest secondary or exploratory.
- For a family of primary tests, adjust the p-values:
  - **Holm:** familywise error rate.
  - **Benjamini–Hochberg:** false discovery rate.

Missing data: session 8. Many outcomes and summary indices: session 10. Lin & Green (2016), *Standard Operating Procedures: A Safety Net for Pre-Analysis Plans*, PS 49(3).



Bring your analysis plan to the workshop

- **Estimand:** the effect on which population of questions, users, or annotators, measured on which outcome.
- **Estimator:** difference in means, blocked or paired, or Lin with named pretreatment covariates.
- **Standard error:** clustered where sampling or assignment groups units; two-way if annotators are sampled.
- **Interval:** with few questions or clusters, a t interval on the CR2 SE or randomization inference. For binary accuracy near 0 or 1, Wilson.
- **Judge-scored outcomes:** the fraction humans label at random, and the correction (DSL or PPI++).



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Your reflection should trace out a step-by-step analytical recipe that you will follow over all relevant components.

